# ANALYSTLAB AFRICA

# DATA ANALYTICS INTERNSHIP

# WEEK 5 PROJECT REPORT

# SwiftLog Logistics Analytics & AI Transformation Project

*Title: Enterprise supply chain Analytics and Artificial Intelligence Transformation for SwiftLog Logistics Limited.*

TOOLS: Microsoft Power BI Desktop (Power Querry and DAX)

**INTERN:** OTUTU LINDA CHINECHEREM

# SwiftLog Logistics Analytics & AI Transformation Project

## Part 5: Business Analysis & Data Storytelling

## Basis of the Analysis

The analysis uses the 11 SwiftLog datasets as prepared in Parts 1 to 3 and shown on the five dashboard pages: Executive, Customers, Inventory & Warehouse, Shipments & Delivery, and Suppliers & Products. Orders run from 1 January to 8 September 2026, so September is a partial month. Figures match the DAX measures in the dashboard. The charts in this document reproduce the dashboard visuals for each question, and the page each one comes from is named in the answer.

**Important notes on what the data can and cannot show:**

- Supplier results rest on a simulated product-to-supplier link, because the source data has no supplier key. They illustrate the method and should be confirmed with real supplier data.
- On time means 2 days of delay or less. Warehouse capacity (100,000 units) and the 3-day delivery lead time are assumptions.
- Returns are recorded per order, so a product return rate is the share of that product's orders that had a return.
- Differences between warehouses and regions are modest. The findings point to where to look first, not to large gaps.

### Headline KPIs

| KPI | Value | KPI | Value |
|---|---|---|---|
| Total Revenue | 21.67 million | On-Time Delivery Rate | 36.3% |
| Net Revenue (excl. cancelled) | 14.44 million | Average Delivery Delay | 3.57 days |
| Total Orders | 5,000 | Return Rate | 9.5% |
| Customers (active) | 1,000 (991) | Cancellation Rate | 32.5% |
| Average Order Value | 4,548 | Inventory Turnover | 0.08 |
| Repeat Customer Rate | 96.0% | Low Stock Records (<100) | 746 (18.7%) |

## Analysis of the Dashboard

### 1. Which products generate the highest revenue?

**Answer:** P040 (Electronics) is the top product with 191,464 in revenue, followed by P085 (Food) with 187,416 and P186 (Fashion) with 180,896. At category level, Fashion leads with 8.06 million (37.2%), then Food with 7.11 million (32.8%) and Electronics with 6.50 million (30.0%).

| Rank | Product | Category | Unit price | Units sold | Revenue |
|---|---|---|---|---|---|
| 1 | P040 | Electronics | 473.92 | 404 | 191,464 |
| 2 | P085 | Food | 493.20 | 380 | 187,416 |

| Rank | Product | Category | Unit price | Units sold | Revenue |
|---|---|---|---|---|---|
| 3 | P186 | Fashion | 461.47 | 392 | 180,896 |
| 4 | P234 | Food | 422.61 | 425 | 179,609 |
| 5 | P125 | Fashion | 395.23 | 442 | 174,692 |
| 6 | P025 | Food | 464.12 | 374 | 173,581 |
| 7 | P209 | Fashion | 478.87 | 357 | 170,957 |
| 8 | P220 | Food | 453.88 | 372 | 168,843 |
| 9 | P035 | Food | 400.96 | 421 | 168,804 |
| 10 | P100 | Fashion | 423.97 | 393 | 166,620 |

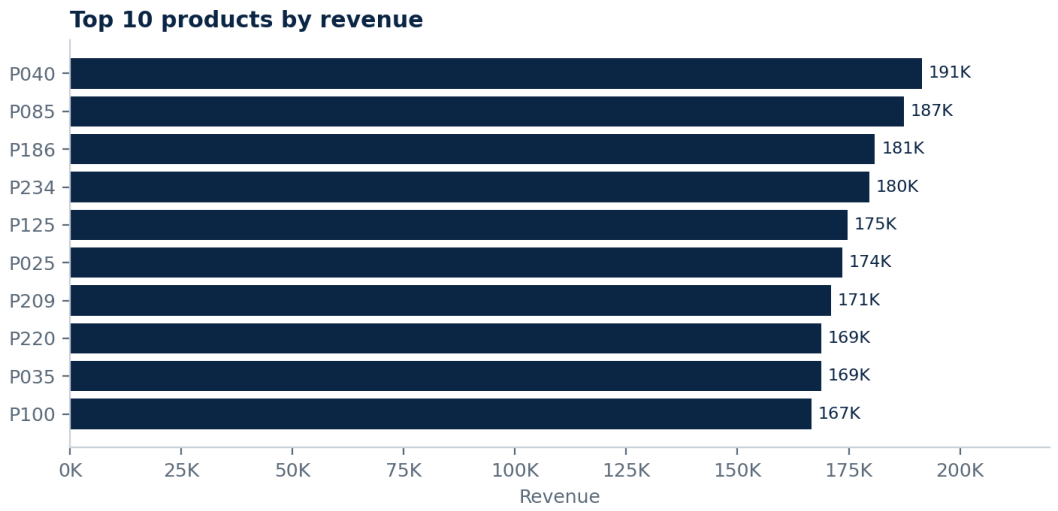



*Figure 1: Top 10 products by revenue (Suppliers & Products page)*

**Business meaning:** Revenue is spread widely. The top 10 products make up only 8.1% of revenue and the top 25 make up 19.2%, so there is no heavy dependence on one product. Every top-25 product has a unit price of 395 or more, which shows that revenue is driven by high price points as much as by volume.

## 2. Which customers contribute the highest revenue?

**Answer:** Customer 263 (Lagos) contributes the most at 74,103, followed by Customer 815 (Ibadan) at 70,177 and Customer 512 (Kano) at 69,852. By city, Kano generates the most revenue at 5.68 million (26.2%).

| Rank | Customer | City | Orders | Revenue |
|---|---|---|---|---|
| 1 | Customer 263 | Lagos | 9 | 74,103 |
| 2 | Customer 815 | Ibadan | 12 | 70,177 |
| 3 | Customer 512 | Kano | 10 | 69,852 |
| 4 | Customer 223 | Kano | 8 | 69,398 |
| 5 | Customer 597 | Lagos | 9 | 65,425 |
| 6 | Customer 924 | Lagos | 9 | 64,158 |

| Rank | Customer | City | Orders | Revenue |
|---|---|---|---|---|
| 7 | Customer 581 | Ibadan | 15 | 62,453 |
| 8 | Customer 300 | Lagos | 11 | 61,745 |
| 9 | Customer 30 | Kano | 10 | 60,863 |
| 10 | Customer 489 | Kano | 11 | 60,159 |

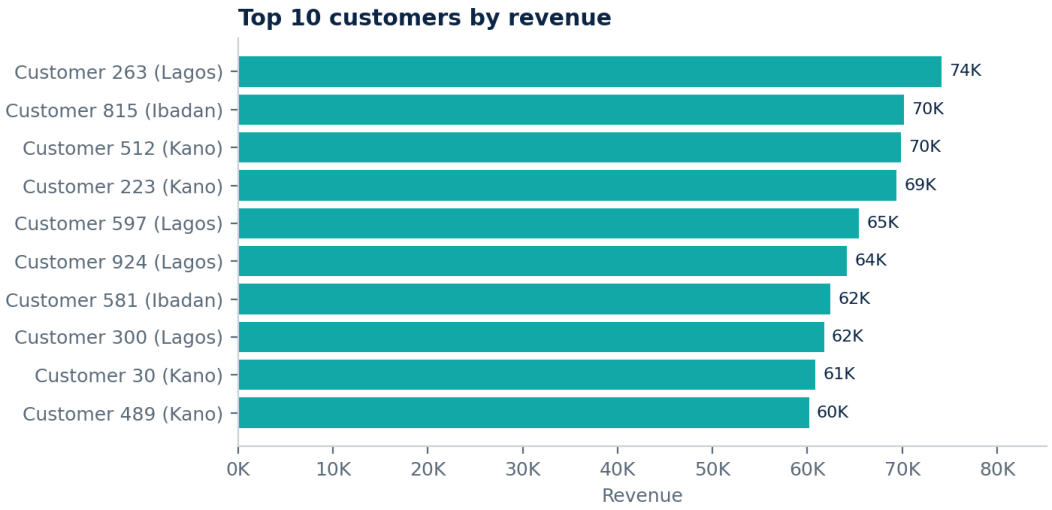



*Figure 2: Top 10 customers by revenue (Customers page)*

**Business meaning:** The customer base is broad. The top 10 customers provide 3.0% of revenue and the top 100 provide 22.1%. City revenue is also balanced: Kano 26.2%, Lagos 25.5%, Abuja 24.2% and Ibadan 24.1%. Customer 581 placed the most orders (15).

## 3. Which warehouses operate most efficiently?

**Answer:** Measured by on-time delivery and average delay, W01 (Abuja) is the most efficient with 41.3% on time and a 3.46-day average delay. W10 (Kano) follows with 40.1% and W07 (Lagos) with 39.6%. W05 (Lagos), W09 (Lagos) and W03 (Abuja) are the weakest at 32.0%, 32.5% and 32.6%.

| Warehouse | Location | Shipments | On-time rate | Avg delay (days) | Shipments per employee |
|---|---|---|---|---|---|
| W01 | Abuja | 332 | 41.3% | 3.46 | 22.1 |
| W10 | Kano | 334 | 40.1% | 3.49 | 22.3 |
| W07 | Lagos | 371 | 39.6% | 3.47 | 21.8 |
| W03 | Abuja | 344 | 32.6% | 3.68 | 20.2 |
| W09 | Lagos | 332 | 32.5% | 3.68 | 23.7 |
| W05 | Lagos | 334 | 32.0% | 3.79 | 23.9 |

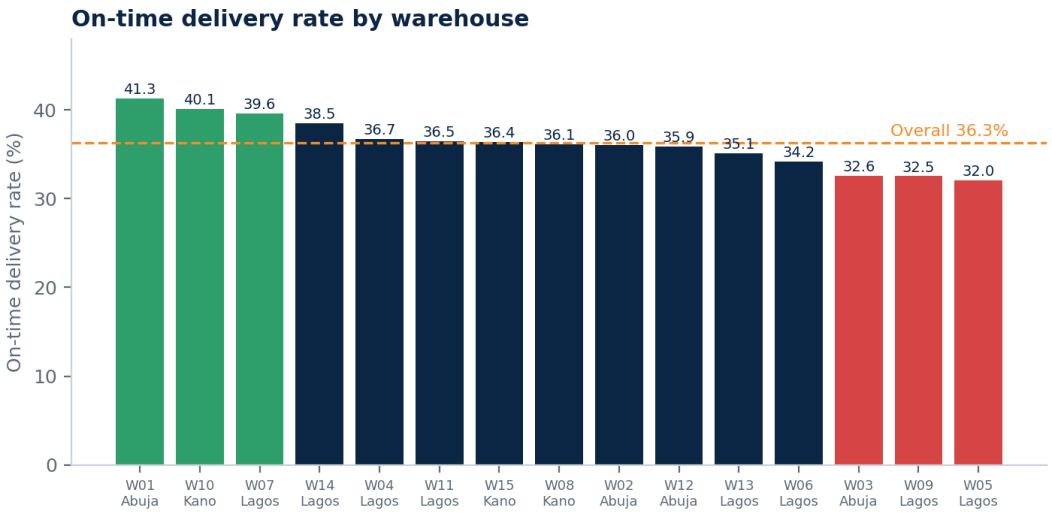



*Figure 3: On-time delivery rate by warehouse (Inventory & Warehouse and Shipments & Delivery pages)*

**Business meaning:** The best and worst warehouses are 9.3 percentage points apart, and none reaches 42%. The top three handle similar volumes to the bottom three, so the gap points to process differences that can be copied, not to workload alone.

## 4. Which suppliers consistently perform well?

**Answer:** Supplier 85 and Supplier 116 are the only suppliers that rank in the top 20 in both halves of the period (January to May and June to September). Supplier 85 has the highest overall score at 67.3. These results use the simulated supplier link described above.

| Supplier | Score | Orders | On-time | Return rate | Cancel rate |
|---|---|---|---|---|---|
| Supplier 85 | 67.3 | 115 | 47.8% | 5.2% | 25.2% |
| Supplier 106 | 66.3 | 60 | 45.0% | 5.0% | 23.3% |
| Supplier 116 | 65.4 | 65 | 52.3% | 6.2% | 44.6% |
| Supplier 141 | 65.4 | 54 | 46.3% | 7.4% | 27.8% |
| Supplier 145 | 65.3 | 57 | 47.4% | 10.5% | 26.3% |
| Lowest: Supplier 134 | 51.6 | 64 | 28.1% | 12.5% | 43.8% |
| Lowest: Supplier 35 | 52.7 | 124 | 28.2% | 9.7% | 42.7% |
| Lowest: Supplier 108 | 52.7 | 60 | 28.3% | 18.3% | 30.0% |

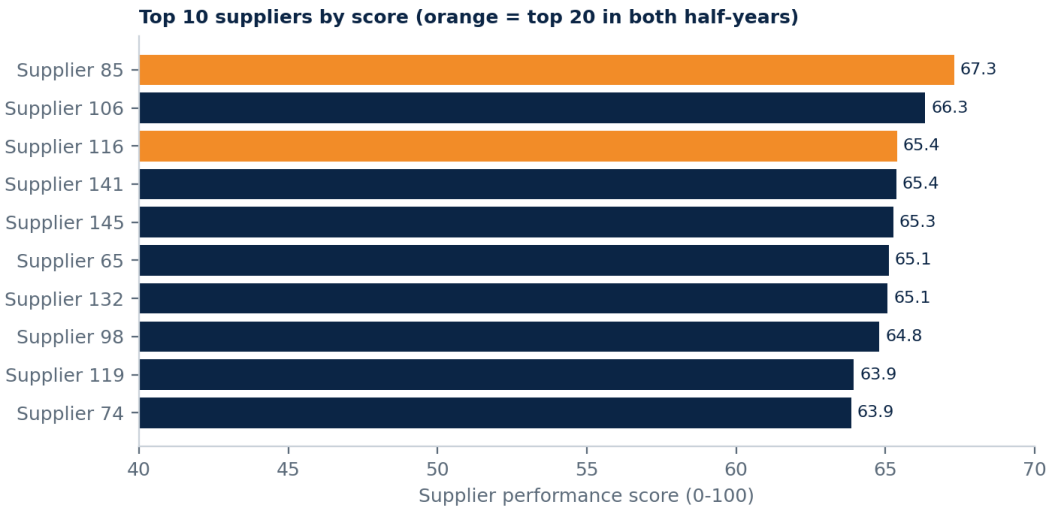



*Figure 4: Top 10 suppliers by performance score (Suppliers & Products page)*

**Business meaning:** Scores range from 51.6 to 67.3 with an average of 59.0. Consistency is rare: the correlation between first-half and second-half scores is only 0.07, so a high score in one period rarely repeats. Supplier 116 holds a high score through strong on-time delivery (52.3%) despite a high cancellation rate, so it needs a closer look.

## 5. Which products have the highest return rates?

**Answer:** Among products with at least 40 orders, P032 has the highest return rate at 21.3% of its orders, then P017 (18.9%), P108 (18.3%), P020 (16.9%) and P177 (16.9%). The overall return rate is 9.5%.

| Product | Orders | Orders with a return | Return rate |
|---|---|---|---|
| P032 | 47 | 10 | 21.3% |
| P017 | 53 | 10 | 18.9% |
| P108 | 60 | 11 | 18.3% |
| P020 | 59 | 10 | 16.9% |
| P177 | 65 | 11 | 16.9% |

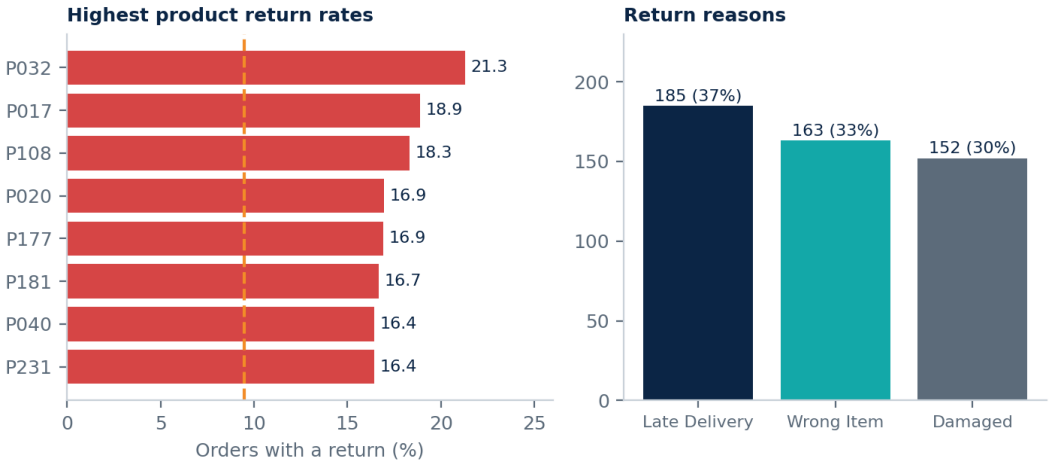



*Figure 5: Highest product return rates and return reasons (Executive and Suppliers & Products pages)*

**Business meaning:** By category, return rates are almost identical: Food 9.8%, Fashion 9.4% and Electronics 9.3%. Product-level rates rest on small samples (about 10 returns each), so treat the product list as a watchlist. The reasons are all operational: Late Delivery (185 returns, 37%), Wrong Item (163, 32.6%) and Damaged (152, 30.4%). P040, the top revenue product, also appears on the high-return list at 16.4%.

## 6. Which regions experience the most delayed deliveries?

**Answer:** By customer city, Kano has the highest share of delayed deliveries (65.7%, or 845 shipments), followed by Abuja (64.4%), Ibadan (63.8%) and Lagos (61.0%). By warehouse location, Lagos has the most delayed shipments (1,701) and the highest delay rate (64.3%).

| Customer city | Shipments | Delayed | Delayed share |
|---|---|---|---|
| Kano | 1,287 | 845 | 65.7% |
| Abuja | 1,205 | 776 | 64.4% |
| Ibadan | 1,240 | 791 | 63.8% |
| Lagos | 1,268 | 774 | 61.0% |

| Warehouse location | Shipments | Delayed | Delayed share |
|---|---|---|---|
| Lagos (8 warehouses) | 2,647 | 1,701 | 64.3% |
| Abuja (4 warehouses) | 1,335 | 849 | 63.6% |
| Kano (3 warehouses) | 1,018 | 636 | 62.5% |

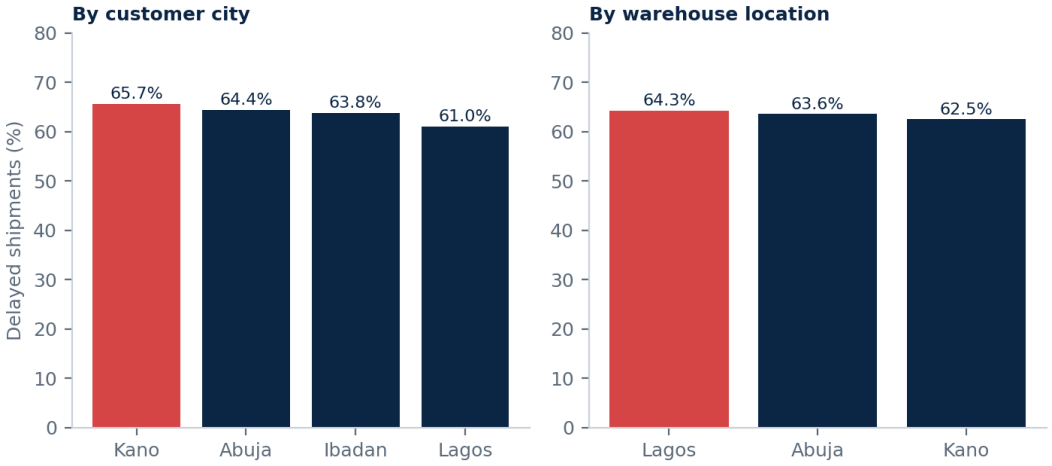



*Figure 6: Delayed shipments by customer city and warehouse location (Shipments & Delivery page)*

**Business meaning:** Delays are widespread, with every region above 60%. Kano customers are the most affected, while Lagos-based warehouses carry 52.9% of all shipments and so account for the largest number of delays.

## 7. Which months record peak business activity?

**Answer:** August is the peak month with 654 orders and 2.85 million in revenue, followed by April (638 orders, 2.74 million) and May (612 orders, 2.72 million). September is partial (to 8 September) and cannot be compared.

| Month | Orders | Revenue (million) | On-time rate |
|---|---|---|---|
| January | 594 | 2.58 | 36.7% |
| February | 577 | 2.51 | 36.9% |
| March | 597 | 2.60 | 38.9% |
| April | 638 | 2.74 | 33.5% |
| May | 612 | 2.72 | 37.4% |
| June | 579 | 2.55 | 37.3% |
| July | 595 | 2.54 | 36.0% |
| August | 654 | 2.85 | 35.8% |
| September (1 to 8) | 154 | 0.59 | 28.6% |

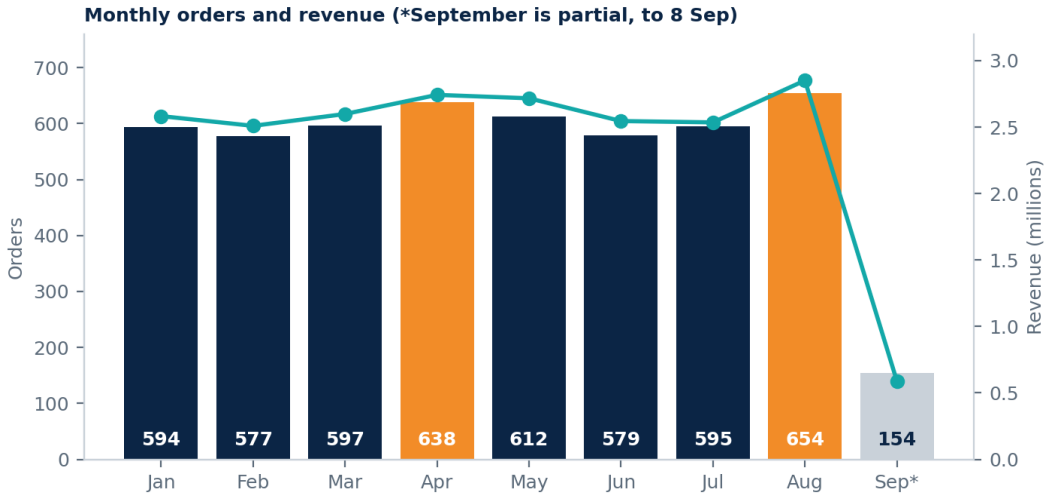



*Figure 7: Monthly orders and revenue (Executive page)*

**Business meaning:** Demand is steady at 577 to 654 orders per full month, with peaks in April and August. April also had the weakest service of any full month (33.5% on time), so peak periods put delivery under the most pressure.

## 8. Which KPIs require immediate management attention?

**Answer:** On-time delivery (36.3%) and order cancellations (32.5%) need action first, followed by the pending-order backlog and inventory turnover.

| KPI | Current value | Why it needs attention |
|---|---|---|
| On-Time Delivery Rate | 36.3% | 63.7% of shipments (3,186) are delayed by more than 2 days |
| Average Delivery Delay | 3.57 days | 1,297 shipments (25.9%) are delayed by 6 to 7 days |
| Cancellation Rate | 32.5% | 1,626 orders cancelled, worth 7.23 million |
| Delivered orders | 33.2% | Only 1,658 of 5,000 orders are Delivered; 1,716 remain Pending |
| Pending backlog | 1,270 orders | About 1,270 pending orders are older than 60 days, worth 5.54 million |
| Inventory Turnover | 0.08 | Stock is about 12 times the units ordered |

| KPI | Current value | Why it needs attention |
|---|---|---|
| Low Stock Records | 746 (18.7%) | Records below 100 units, including 9 at zero |
| Return Rate | 9.5% | All reasons are operational and preventable |

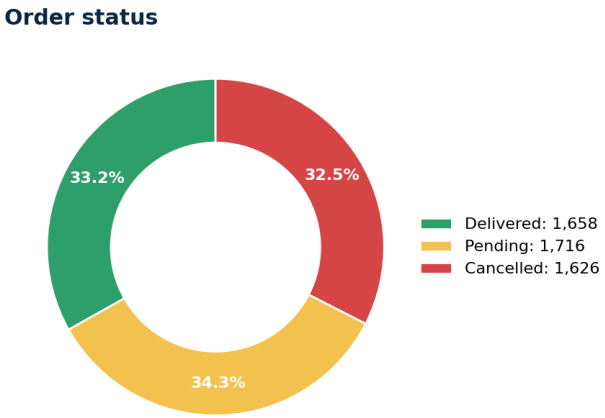



*Figure 8: Order status (Executive page)*

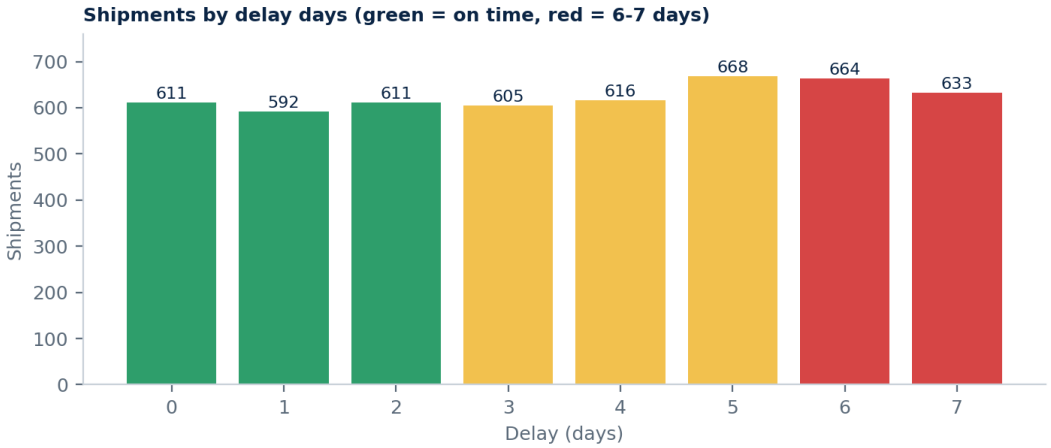



*Figure 9: Shipments by delay days (Shipments & Delivery page)*

## 9. What operational inefficiencies were identified?

- Widespread late delivery: 63.7% of shipments are late and every warehouse is below 42% on time.
- Order fulfilment leakage: only one in three orders is delivered, while a third is cancelled and a third is pending. Pending and cancelled orders each hold more than 7 million in value.
- Stale pending orders: about 1,270 orders have waited more than 60 days.
- Inventory imbalance: 1,006,935 units are in stock against 82,442 units ordered, yet 18.7% of stock records are below 100 units. Stock is in the wrong places, not just in excess.
- Uneven workload: shipments per employee range from 19.9 at W13 to 37.8 at W14.
- Preventable returns: late delivery, wrong items and damaged goods account for all 500 returns.
- Data quality: 236 orders have no order lines, 335 of 500 returns belong to Cancelled or Pending orders, and Inventory held duplicate and repeated stock records.

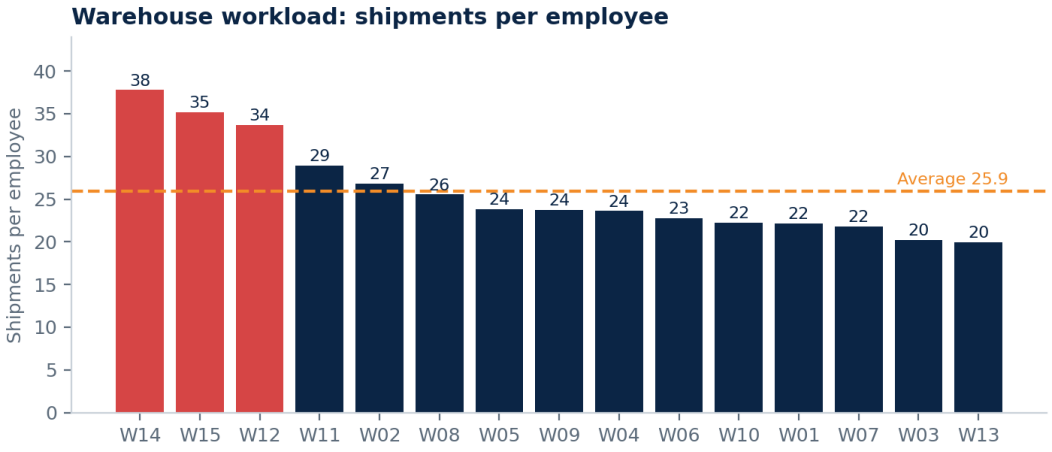



*Figure 10: Shipments per employee by warehouse (Inventory & Warehouse page)*

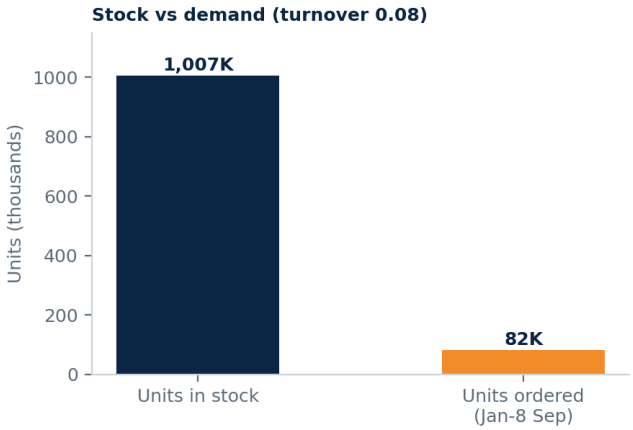



*Figure 11: Units in stock compared with units ordered (Inventory & Warehouse page)*

## 10. What business opportunities should SwiftLog pursue?

- Recover pending revenue: pending orders are worth 7.33 million, and 5.54 million of it is more than 60 days old. Each 10% recovered adds about 0.55 million.
- Reduce cancellations: lowering the cancellation rate by 5 percentage points would keep about 250 orders, roughly 1.1 million at the average order value of 4,548.
- Grow the premium mix: products priced at 300 or more produce 63.3% of revenue, while budget products produce only 3.5%.
- Build on loyalty: 96.0% of customers have ordered more than once and the average is 5.05 orders per customer. Loyalty offers can protect this base. Reactivation can reach the 40 one-time customers and the 9 who have never ordered.
- Plan for peaks: April and August are the busiest months, so capacity can be scheduled ahead.
- Copy what works: W01, W10 and W07 deliver the best on-time rates and can model practices for W05, W09 and W03.
- Use Kano as a growth market: it has the highest revenue share (26.2%) but the highest delay rate (65.7%), so improving service there protects the biggest market.

## Summary

The analysis shows a business with strong, broad demand and loyal customers whose results are held back by execution. Delivery reliability (36.3% on time), order completion (only one in three orders delivered) and inventory placement are the areas that need management attention. The findings feed directly into the recommendations in Part 6.